Exercise 4

Consider split conformal prediction with calibration data set $(X_1, Y_1), \cdots, (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$ at given level $\alpha \in (0, 1)$ and given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. We look at the aggregating prediction sets by the majority vote $\mathcal{C}^{\text{mv}}(X_{n+1})$ of two conformal algorithms $\mathcal{C}_1$ and $\mathcal{C}_2$, where $\mathcal{C}_1$ uses s as score and $\mathcal{C}_2$ uses $-s$ as score. Give an explicit formula of $\mathcal{C}^{\text{mv}}(X_{n+1})$ and its marginal coverage level, assuming exchangeability of the data set, symmetry of s, and almost surely no ties in S_i's.

Solution We first compute the expression of each conformal set. Let us use the order statistics $S_{(1)} < \cdots < S_{(n)}$ of $S_1, \cdots, S_n$. Put

$$k = \lceil (n+1)(1-\alpha) \rceil.$$

We discuss by cases of whether $k = n + 1$.

First of all, in the usual case, $k < n + 1$. The rule $(-v)_{(i)} = -v_{(n-i+1)}$ for any $v \in \mathbb{R}^n$ tells us that, by definition of $\mathcal{C}_1$ and $\mathcal{C}_2$:

$$\mathcal{C}_1(x) = \left\{ y \in \mathcal{Y} \mid s(x, y) \leqslant S_{(k)} \right\}$$

and

$$\mathcal{C}_2(x) = \left\{ y \in \mathcal{Y} \mid s(x, y) \geqslant S_{(n+1-k)} \right\}.$$

Thus, by definition:

$$\mathcal{C}^{\text{mv}}(x) = \left\{ y \in \mathcal{Y} \mid \frac{1}{2} \sum_{i=1}^{2} \chi_{y \in C_i(x)} > \frac{1}{2} \right\} = \mathcal{C}_1(x) \cap \mathcal{C}_2(x) = \left\{ y \in \mathcal{Y} \mid S_{(n+1-k)} \leqslant s(x, y) \leqslant S_{(k)} \right\}.$$

Next, for the trivial case $k = n + 1$, we have $\mathcal{C}^{\text{mv}}(x) = \mathcal{C}_1(x) = \mathcal{C}_2(x) = \mathcal{Y}$. Therefore:

$$\mathcal{C}^{\text{mv}}(X_{n+1}) = \begin{cases} \left\{ y \in \mathcal{Y} \mid S_{(n+1-k)} \leqslant s(X_{n+1}, y) \leqslant S_{(k)} \right\} & k = \lceil (n+1)(1-\alpha) \rceil < n+1, \\ \mathcal{Y} & k = n+1. \end{cases}$$

A simple counting of the probability of placing S_{n+1} gives

$$P\left\{ Y_{n+1} \in \mathcal{C}^{\text{mv}}(X_{n+1}) \right\} = \frac{\max\{0, 2k - n - 1\}}{n+1} = \max\left\{ 0, 1 - \frac{2\lfloor \alpha(n+1) \rfloor}{n+1} \right\}.$$

It just happens that this formula works for $k = n + 1$ case as well. This completes the computation. ////